

**Sri Lanka Institute of Information
Technology**



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Web-Based Supermarket Ordering System

Name	Student ID Number
Jayalath J.G.H	IT24101355
Alahakoon B.M.B.S	IT24101474
Dinod P.A.S	IT24101334
Sanvidu M.G.M	IT24101458
Fernando S.A.W	IT24101304
Thahani N	IT24104344

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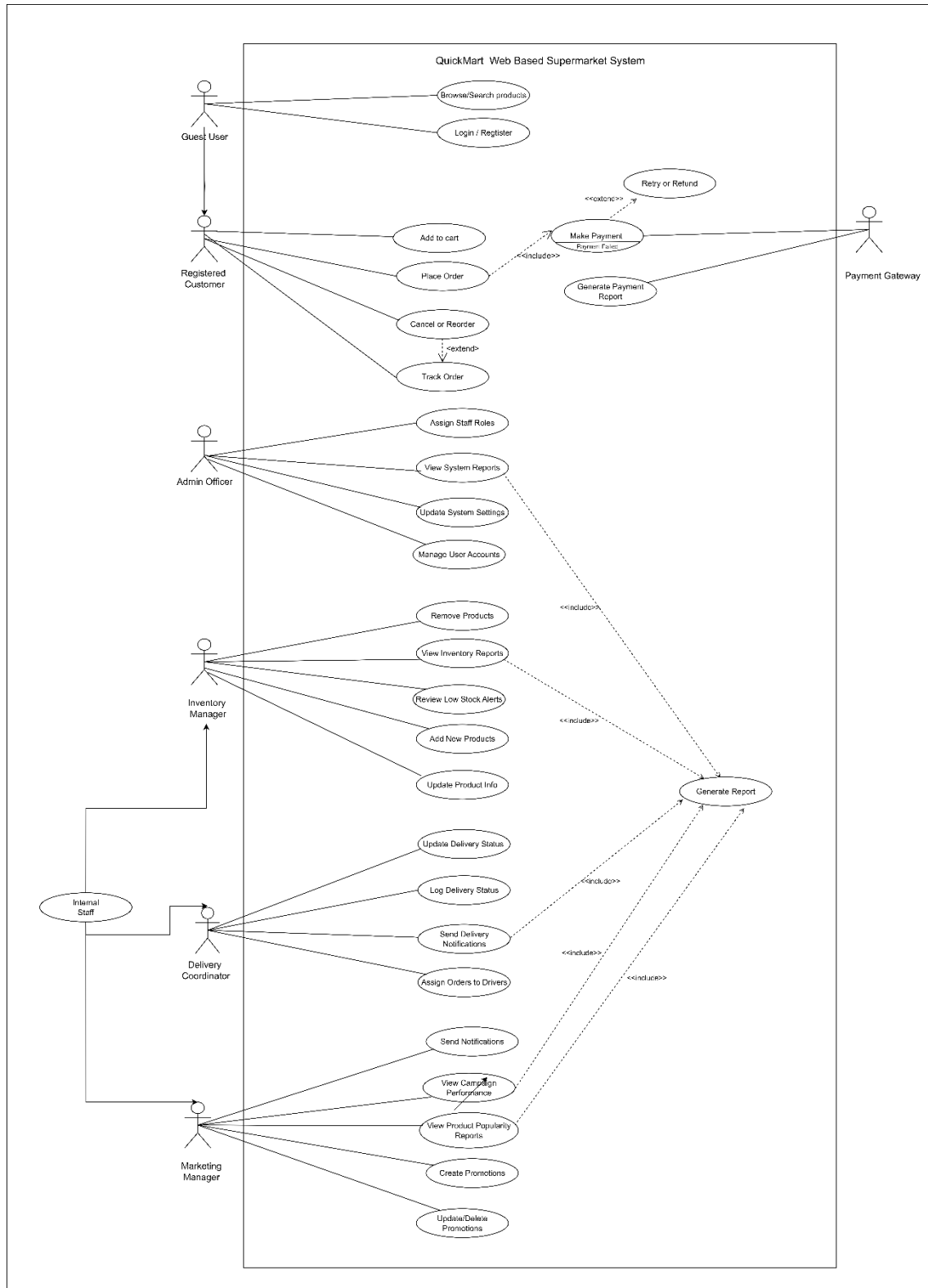
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Contents

1. Use Case Diagram.....	3
2. Use Case Scenarios	4
1. User Management.....	4
2. Product and Inventory Management.....	5
3. Order Processing System.....	6
4. Payment Management	7
5. Delivery Management	8
6. Marketing & Promotion Management.....	9

1. Use Case Diagram



2. Use Case Scenarios

1. User Management

- Jayalath J.G.H. (IT24101355)

Field	Description
Number	1
Name	Manage Users & Permissions
Summary	Admin Officer manages user accounts, assigns roles, and monitors activities.
Priority	High
Preconditions	Admin must be logged in with valid credentials.
Postconditions	Users and roles updated; system integrity maintained.
Primary Actor(s)	Admin Officer
Secondary Actor(s)	System Database
Trigger	Admin logs in.
Main Scenario	1. Admin logs in 2. Views dashboard 3. Creates/updates/deletes user accounts 4. Assigns roles 5. Reviews reports 6. Suspends/reactivates accounts
Extensions	Invalid user details → <<extends>> Display error message
Includes	<<includes>> Assign Roles
Open Issues	None

- Outcome:** Secure and organized user management with clear role-based access.

2. Product and Inventory Management

- Alahakoon B.M.B.S (IT24101474)

Field	Description
Number	2
Name	Manage product and inventory
Summary	Inventory Manager updates product catalog and stock to ensure accuracy and availability.
Priority	High
Preconditions	Inventory manager must be logged into the proper
Postconditions	Product catalog and stock levels are updated
Primary Actor(s)	Inventory Manager
Secondary Actor(s)	System, Suppliers
Trigger	Low stock alerts
Main Scenario	1. Manager logs into the system. 2. System displays product catalog and stock status. 3. Checks current stock levels. 4. Adds or updates product details. 5. System generates alerts for low-stock items.
Extensions	Stock data unavailable →<<extends>> System notifies manager of error.
Includes	<<includes>> Check current stock
Open Issues	None

- Outcome:** Accurate inventory and up-to-date product catalog.

3. Order Processing System

- Fernando S.A.W (IT24101304)

Field	Description
Number	3
Name	Manage Orders
Summary	Customer browses products, places orders, and tracks them.
Priority	High
Preconditions	Customer must be registered and logged in.
Postconditions	Order successfully placed, tracked, or canceled.
Primary Actor(s)	Regular Customer
Secondary Actor(s)	Admin Officer, Delivery Coordinator
Trigger	Customer logs in.
Main Scenario	1. Customer logs in 2. Browses/searches products 3. Adds to cart/wish-list 4. Places order 5. Cancels or reorders as needed 6. Tracks order status
Extensions	Out of stock → <<extends>> Suggest Alternatives
Includes	<<include> Confirm Order
Open Issues	None

- **Outcome:** Customer can shop, order, and track purchases smoothly.

4. Payment Management

- Sanvidu M.G.M (IT24101458)

Field	Description
Number	4
Name	Process Payments
Summary	Payment Manager ensures secure payment handling and reporting.
Priority	High
Preconditions	Customer must have placed an order.
Postconditions	Payment confirmed; receipt generated.
Primary Actor(s)	Payment Manager
Secondary Actor(s)	Customer, Payment Gateway
Trigger	Customer initiates payment.
Main Scenario	1. Customer selects payment method 2. System redirects to payment gateway 3. Payment processed (success/fail) 4. Receipt generated 5. Payment Manager reviews logs and reports
Extensions	Payment failed → <<extends>> Retry/Refund
Includes	<<includes>> Generate Receipt
Open Issues	None

- **Outcome:** Payment is securely processed, and payment status is updated accordingly.

5. Delivery Management

- Dinod P.A.S (IT24101334)

Field	Description
Number	5
Name	Manage Deliveries
Summary	Delivery Coordinator assigns and monitors deliveries.
Priority	Medium
Preconditions	Confirmed orders exist.
Postconditions	Deliveries completed and confirmed.
Primary Actor(s)	Delivery Coordinator
Secondary Actor(s)	Drivers, Customers
Trigger	Coordinator logs in.
Main Scenario	1. Coordinator logs in 2. Views pending orders 3. Assigns delivery staff 4. Updates status (packed, in transit, delivered) 5. Sends notifications 6. Logs delivery reports
Extensions	Location not found → <<extends>> Check Alternative Route
Includes	<<includes>> Assign Drivers
Open Issues	None

- **Outcome:** Orders are delivered on time with proper tracking and issue handling.

6. Marketing & Promotion Management

- Thahani N (IT24104344)

Field	Description
Number	6
Name	Manage Promotions & Campaigns
Summary	Marketing Manager handles campaigns, discounts, and notifications.
Priority	Medium
Preconditions	Marketing Manager must be logged in.
Postconditions	Campaigns created and customer engagement increased.
Primary Actor(s)	Marketing Manager
Secondary Actor(s)	Customers, System Database
Trigger	Manager logs in.
Main Scenario	1. Manager logs in 2. Creates promotions/campaigns 3. Updates discounts 4. Sends personalized notifications 5. Reviews campaign performance
Extensions	Invalid campaign → <<extends>> Notify Admin
Includes	<<includes>> Send Notifications
Open Issues	None

- **Outcome:** Increased customer engagement and improved sales through promotions.